



OPEN SOURCE CODING ASSIGNMENT PART 1

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1. Introduction

1.1 Overview of the Application

This document outlines the planning and design of a personal budgeting mobile application called **SmartBudget Pro**, developed for the Android platform. The purpose of the application is to help users effectively manage their finances by tracking income, monitoring expenses, and setting realistic budgets.

The design of this application is informed by research conducted on three existing budgeting applications: Goodbudget Planner, Monefy (Money Tracker), and Money Manager Expense & Budget. The research identified key strengths such as ease of use, visual reporting, and structured budgeting systems. These insights have directly influenced the design of SmartBudget Pro. (Dayspring Technologies, 2023; Reflectly ApS, 2023; Realbyte Inc., 2023)

The application aims to combine simplicity with advanced functionality, offering both beginner-friendly tools and deeper financial insights.

1.2 App Icon Design

The SmartBudget Pro app icon will include:

- A **wallet symbol**, representing financial management
- An **upward arrow or graph**, symbolizing financial growth
- A **green and blue colour scheme**:
 - Green → money and growth
 - Blue → trust and stability (Lidwell, Holden & Butler, 2010)

1.3 Innovative Features

Based on research findings, the following innovative features will be included:

- **Envelope budgeting system**, inspired by Goodbudget. (Dayspring Technologies, 2023)
- **One-click expense entry** for fast tracking, inspired by Monefy. (Reflectly ApS, 2023)
- **Advanced analytics and reports**, inspired by Money Manager. (Realbyte Inc., 2023)
- **Gamification system** (badges, points, rewards)
- **Goal tracking and savings planner**
- **Smart notifications and reminders**

These features are designed to improve usability, engagement, and financial awareness. (Nielsen, 1994).

2. Application Requirements

2.1 Functional Requirements

User Registration and Login

- Users must be able to register using an email and password
- Secure login functionality must be implemented
- User data must be stored securely and privately
- Passwords should be encrypted for safety

Income and Expense Tracking

- Users must be able to:
 - Add income (amount, date, category)
 - Record expenses quickly and efficiently
- The app must support:
 - One-click entry for frequently used categories
 - Editing and deleting transactions
- All financial data must be stored persistently

Budget Creation (Envelope System)

- Users must be able to:
 - Create budget categories (envelopes)
 - Allocate money to each category
- The app must:
 - Track remaining balances per category
 - Display warnings when budgets are exceeded

Visual Reports and Analytics

- The app must generate:
 - Graphs showing spending distribution
 - Bar graphs for comparisons

- Reports must include:
 - Daily, weekly, and monthly views
- Overspending must be clearly highlighted
-

Goal Setting

- Users must be able to:
 - Set savings goals
 - Track progress visually
- The app must:
 - Notify users when milestones are reached

Notifications and Reminders

- The app must:
 - Send reminders for bills and payments
 - Alert users when budgets are exceeded
- Notifications must be customizable

Data Backup and Synchronisation

- Users must be able to:
 - Back up their data
 - Sync data across devices
- Cloud storage integration may be used

Security

- The app must provide:
 - Password or PIN protection
 - Data encryption for sensitive information

2.2 Non-Functional Requirements

- The app must be **user-friendly and intuitive**. (Nielsen, 1994)

- Performance must be **fast with minimal delays**.
- The interface must be **visually appealing and responsive**.
- The app must be **compatible with Android devices**.
- Strong **data privacy and security** must be ensured. (OWASP, 2021)

2.3 Additional Features

To enhance user experience, the following features will be included:

- **Gamification system** (points, badges, achievements)
- **Dark mode** for improved usability
- **Multi-currency support**
- **Export reports to PDF or Excel**

3. User Interface Design

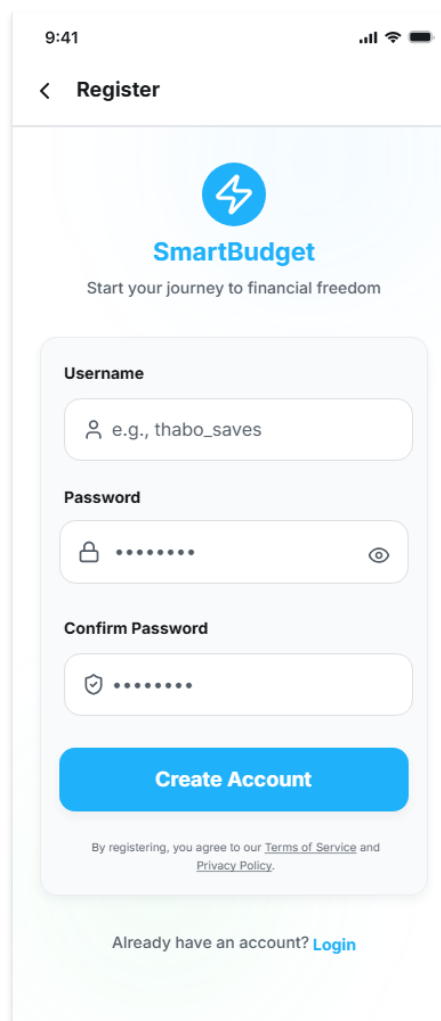
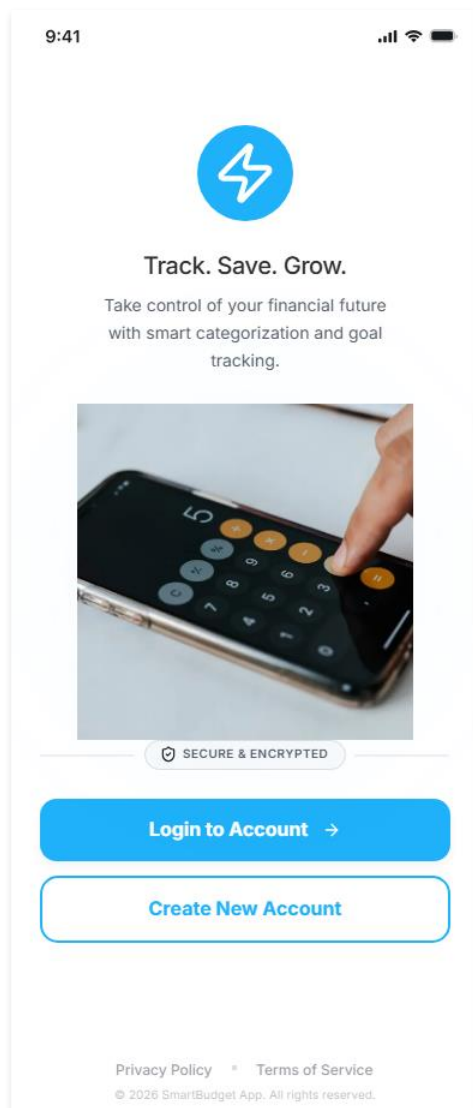
3.1 Screen Descriptions

1. Login/Register Screen

Purpose: Allows users to create an account or log in

Design:

- Email and password input fields
- Login and register buttons

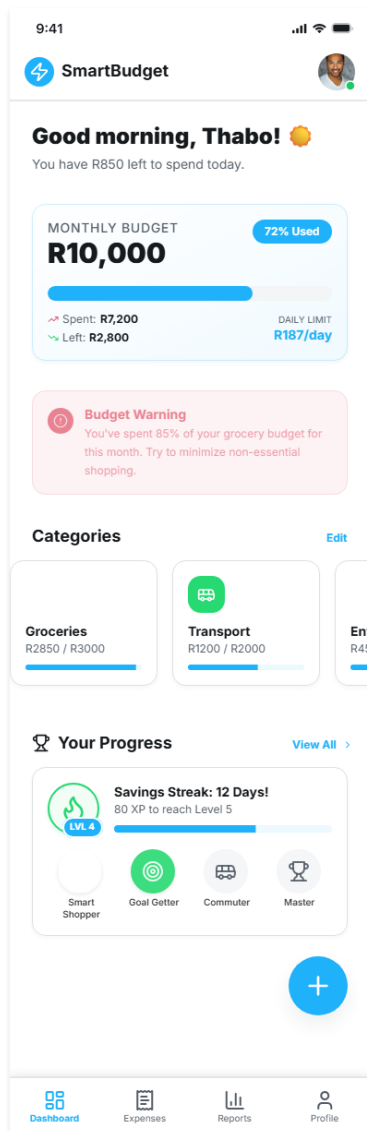


2. Dashboard Screen

Purpose: Displays an overview of financial status

Design:

- Total balance at the top
- Summary of income and expenses
- Pie chart showing spending distribution



3. Add Transaction Screen

Purpose: Allows users to record income or expenses

Design:

- Input fields (amount, category, date)
- Quick-add buttons for common expenses

The image shows a mobile app interface for adding an expense. At the top, the status bar shows the time 9:41 and signal strength. Below the status bar is a header with a back arrow and the text "Add Expense". The main content area is divided into several sections:

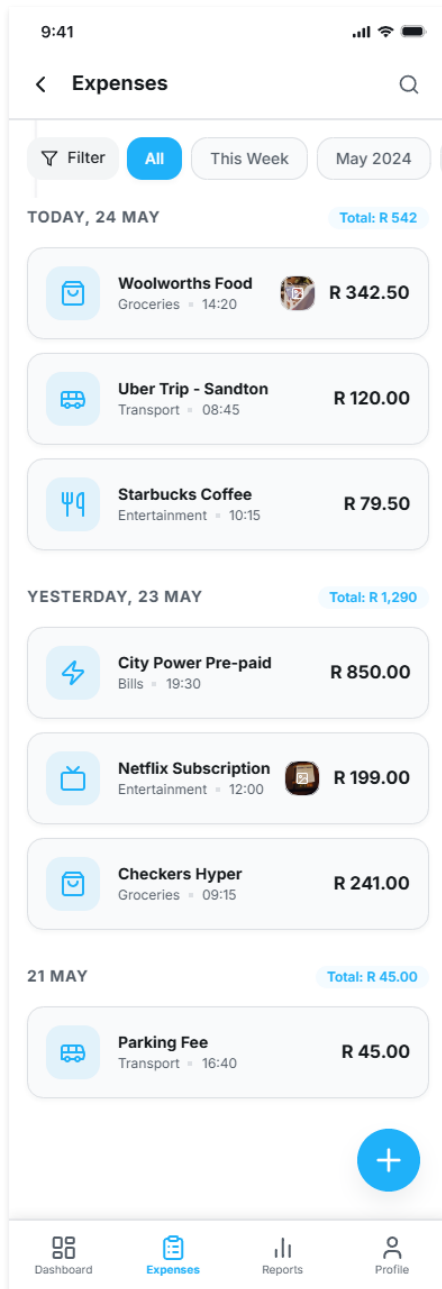
- Amount Section:** A light blue rounded rectangle containing the text "HOW MUCH?" in small blue letters. Below this is a large "R 0.00" in blue. At the bottom of this section, it says "Remaining budget: R 2,450.00" in a smaller font.
- Category Section:** A white rounded rectangle with a light blue border. It contains a category icon (a tag) on the left, the word "CATEGORY" in small blue letters, and a green pill-shaped button with the word "Groceries" in white. A right arrow is on the far right.
- Date Section:** A white rounded rectangle with a light blue border. It contains a calendar icon on the left, the word "DATE" in small blue letters, and the text "Today, 12 June 2024". A right arrow is on the far right.
- Description Section:** A white rounded rectangle with a light blue border. It contains a list icon on the left, the word "DESCRIPTION" in small blue letters, and the text "What was this for?".
- Proof of Purchase Section:** A section with the title "Proof of Purchase" in bold. Below it is a dashed border containing a camera icon, the text "Upload Receipt Photo" in blue, and "JPG, PNG up to 5MB" in small grey letters.
- Save Button:** A large blue rounded rectangle at the bottom with the text "Save Expense" in white.

4. Budget (Envelope) Screen

Purpose: Displays budgeting categories

Design:

- List of envelopes (e.g., food, transport)
- Progress bars showing remaining budget

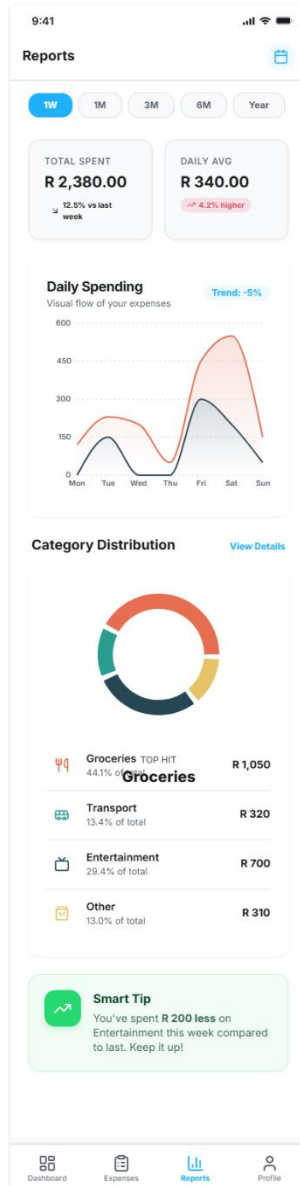


5. Reports Screen

Purpose: Shows financial analytics

Design:

- Graphs and charts
- Filters for time periods



6. Goals Screen

Purpose: Tracks savings goals

Design:

- List of goals
- Progress indicators

7. Settings Screen

Purpose: Allows customization

Design:

- Notification settings

- Theme selection (light/dark mode)
- Backup options

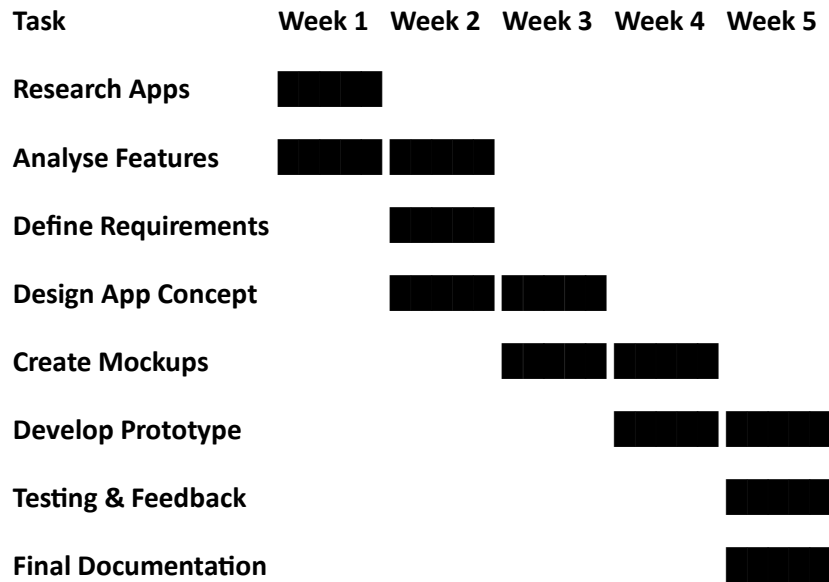
3.2 Navigation Diagram

The navigation flow is as follows:

- Login Screen → Dashboard
- Dashboard → Add Transaction
- Dashboard → Budget Screen
- Dashboard → Reports Screen
- Dashboard → Goals Screen
- Dashboard → Settings

Users can return to the Dashboard from any screen using a navigation bar.

4. Gantt Chart



5. Conclusion

This document has presented the full planning and design of the SmartBudget Pro application. The design is based on research into existing budgeting applications and incorporates their strengths while improving usability and functionality.

By combining features such as envelope budgeting, quick expense tracking, and advanced analytics, the application provides a comprehensive financial management solution. Additional features like gamification and goal tracking further enhance user engagement.

This detailed design ensures that the development process can be carried out efficiently and effectively. (Deterding et al., 2011).

6. References

Dayspring Technologies. (2023). *Goodbudget: Budget Planner*. Available at: <https://goodbudget.com> (Accessed: 25 March 2026).

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